Furkan Sönmez

Date of birth: 25/07/1994 | **Nationality:** Turkish | **Phone number:** (+49) 15236717807 (Mobile) |

Email address: s.furkan@gmail.com | **LinkedIn:** linkedin.com/in/frknsonmez |

Address: Schifferlstraße 6, 80687 , München, Germany (Home)

ABOUT ME

~5 years experienced embedded software engineer specializing in Real-Time safety-critical contexts, particularly in compliance with RTCA DO-178. Proficient in aviation battery topologies.Solid background in Electronics. Adept at communication protocols including ARINC-429, UART, SPI, CAN, I2C, RS485. Experience in Static and Dynamic Software Testing. Skilled in developing software high and low-level requirements in DOORS. Expertise extends to Model-Based Battery SW Development in MatLab/Simulink, documentation preparation and utilization of software development process tools (SVN, GIT, Jira). Versatile collaborator with multidisciplinary teams, including Hardware, Mechanical, and Chemical Engineering. Proven success in collaboration with Certification, Qualification, Safety, and Software Assurance teams in SOI1, SOI2 and SOI3. Proficient in embedded software development for Infineon Tricore, STM32 ARM, Atmel/MicroChip AVR and TI's C2000 architectures. Having SW on things fling or helping other things to fly in the field.

EDUCATION AND TRAINING

09/2013 – 12/2019 Ankara, Türkiye
ELECTRICAL ELECTRONICS ENGINEER Middle East Technical University

Address 06800, Ankara, Türkiye

WORK EXPERIENCE

02/2024 – CURRENT Munich, Germany
EMBEDDED SYSTEMS SOFTWARE DEVELOPER CAPGEMINI ENGINEERING

-Unit Test for Battery Management Software in Gtest
-Model Based System Development with SysML/UML in Magic Draw / Cameo
-Embedded Software Development in Infineon Aurix Tricore for HW verification
-AUTOSAR Classic Training

03/2021 – 12/2023 Ankara, Türkiye
EMBEDDED SYSTEMS SOFTWARE DEVELOPER ASPILSAN ENERGY

- Aviation Battery Management System Design (DAL-A)
- 28Vdc - 270Vdc Low and High Voltage Emergency Battery Pack Design for DO-311
- Knowledge of DO-178C, DO-254, MIL-STD-704F, SAE AS50881, ARP4754, MISRA-C
- Software development in C/C++ , Flashing and Debugging for TI C2000 and STM32 MCUs
-Creation of High and Low Level Requirements in DOORS
- GUI design in Python with Qt
- Construct State of Charge, State of Health, Thermal Management, Protections, Communication algorithms.
- Defining DO178C SW Planning , Development, Verification Process in documentation and SOI-1,2 and 3
- Hands-on Experience in ARINC-429, UART, SPI, CAN, I2C, and RS485 Communications.
- Conducting Code/Model Reviews, Static Code Analysis (LDRA), Coverage Analysis (LDRA) early WCET analysis.
- Modeling SysML/UML in Matlab/Simulink.
-Collaboration with Certification, Qualification, Safety, and Software Assurance teams for SOI Activities.
- Software development process tools (or equivalents) such as SVN, GIT, Doxygen, and Jira.
- Comprehensive use of laboratory equipment i.e. Oscilloscopes, Logic Analyzer, Power Supply, Signal Generator.

09/2019 – 07/2020 Ankara, Türkiye
EMBEDDED SYSTEMS SOFTWARE DEVELOPER EYSA TECHNOLOGY

- Developed embedded software in Ultra-Low Power STM32 ARM / AVR MCUs to achieve 8-year battery life
-Programming on RF SoC in Sub GHz and 2.4 GHz Modules
-Experience in FreeRTOS and Bare-Metal Systems
- Designed, Implemented, and tested custom PCBs for Wireless Fire Detection Systems(upto 1500 IoT devices) , satisfying EN-54 standard including EMI/EMC tests.
- Developed communication algorithms for wired/wireless networking and communications in UART, SPI, I2C, RS232,RS485, USB and WIFI
- Developed embedded software for GSM /GPRS module using TCP/IP.
- Reported and documented technical specifications of prototypes for laboratory tests.
-Comprehensive use of laboratory equipment i.e. Oscilloscopes, Logic Analyzer, Power Supply, Signal Generator.

06/2018 – 07/2018 Istanbul, Türkiye
INTERNSHIP TURKISH AIRLINES - TURKISH TECHNIC

- Internship at Radio Laboratory of Maintenance Department.
- Repair, test, calibration and maintenance of avionic components.

04/2019 – 07/2019 Ankara, Türkiye
CANDIDATE ENGINEER ODOREL IT & SECURITY TECHNOLOGY

- Developed embedded software in AVR MCU for GSM /GPRSmodule using TCP/IP.
- Designed and tested 2-layer PCB in Autodesk Eagle for Police Department
- Reported and documented technical specifications of prototypesfor laboratory tests.

LANGUAGE SKILLS

Mother tongue(s): **TURKISH**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C1	C1	C1	C1	C1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

DIGITAL SKILLS

Programming Languages

C | C++ | Python | Assembly | VHDL

MCU Families

TI C2000 | STM32 | MicroChip

Software

LDRA | Matlab/Simulik | Jira | GIT | SVN | IBM DOORS | Doxygen | Jenkins | Gtest | LT SPICE | Altium

TRAININGS

“RTCA DO178 Software Considereation in Airborne Systems and Equipment Certification” Traning given by Turkish Aerospace Industry in February 2022

“Requirement Management with DOORS” Traning given by Turkish Aerospace Industry in July 2023

“RTCA DO254 Design Assurance Guidance for Airborne Electronic Hardware” Traning given by Turkish Aerospace Industry in Jan 2022

“Introduction to Software Testing” Traning given by Turkish Aerospace Industry in Jun 2023

“Project Technical Management” Traning given by Turkish Aerospace Industry in August 2023

COMMUNICATION AND INTERPERSONAL SKILLS

Cooperative | Result-Oriented | Passionate | Holistic | Interdisciplinary | Outgoing

DRIVING LICENCE

Driving Licence: B | 31/12/2019 – 31/12/2029

HOBBIES AND INTERESTS

Photography | Paragliding | Powerlifter | Travel

https://500px.com/p/sfurkan | P2 Level Paragliding Pilot